

EP2

Exercises from Lectures 5–9

Linear growth, perturbation theory, initial conditions, spherical collapse, and Boltzmann solvers

This sheet collects the exercises, numerical bridges, and validation checks suggested in the lectures and their accompanying notes. Related tasks are grouped once; **extensions and supplementary checks are labelled explicitly**. The required practical scope stated in Lectures 7 and 8 is preserved. No solutions are included; quoted analytic results are validation targets from the lectures.

Conventions. Use $N = \ln a$ and $X_N = dX/dN$ for growth and collapse equations; N_g denotes the mesh size per dimension. Peculiar velocity is $\mathbf{v} = a d\mathbf{x}/dt$, and $\mathcal{H} = aH$. Record units, Fourier normalization, spectrum epoch, and random seeds whenever relevant.

Exercise 1. Compute and validate linear growth

Lecture 5 — numerical workflow

Consider pressureless matter, curvature, and a smooth cosmological constant, neglecting radiation in this late-time model. Define

$$E^2(a) = \Omega_{m0}a^{-3} + \Omega_{k0}a^{-2} + \Omega_{\Lambda0}, \quad H = H_0 E, \quad \Omega_m(a) = \frac{\Omega_{m0}a^{-3}}{E^2(a)}.$$

- (a) Implement the growing-mode equation in N :

$$G_{NN} + \left(2 + \frac{d \ln H}{dN}\right) G_N - \frac{3}{2} \Omega_m(a) G = 0.$$

Start sufficiently early in the matter-dominated regime of this model with $G(a_i) = G_N(a_i) = a_i$. Return $D(a) = G(a)/G(1)$ and $f(a) = G_N/G$, without numerically differentiating an interpolated D .

- (b) Recover $D = a$ and $f = 1$ in Einstein–de Sitter (EdS). Compare with flat Λ CDM, $\Omega_{m0} = 0.3$, and open CDM, $\Omega_{m0} = 0.3$, $\Omega_{\Lambda0} = 0$. Plot $D(a)$ and $f(a)$, always stating the normalization.
- (c) Independently evaluate the quadrature

$$G(a) = \frac{5\Omega_{m0}}{2} E(a) \int_0^a \frac{d\tilde{a}}{\tilde{a}^3 E^3(\tilde{a})}, \quad D(a) = \frac{G(a)}{G(1)},$$

and compare it with the ODE result over the integration interval.

- (d) Vary a_i and the integration tolerances. Report the changes in both D and f , and distinguish initial-mode errors from solver errors.

Source: Lecture 5, slides 38–45; notes, Sections 3.4–3.7.

Exercise 2. Measure the statistics generated by mode coupling

Lecture 6 — numerical bridge

Use the Gaussian-field generator of Exercise 3. Work at a fixed epoch with small linear variance, in a periodic volume $V = L^3$, and preserve the original realization δ_1 .

- (a) Construct the second-order SPT density,

$$\delta_{2,\mathbf{k}} = \frac{1}{V} \sum_{\mathbf{q}} F_2(\mathbf{q}, \mathbf{k} - \mathbf{q}) \delta_{1,\mathbf{q}} \delta_{1,\mathbf{k}-\mathbf{q}},$$

using the symmetric EdS kernel

$$F_2(\mathbf{k}_1, \mathbf{k}_2) = \frac{5}{7} + \frac{\mu}{2} \left(\frac{k_1}{k_2} + \frac{k_2}{k_1} \right) + \frac{2}{7} \mu^2, \quad \mu = \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_2.$$

Document the treatment of zero modes, the mesh band limit, and aliasing. The two input fields already contain their growth at the chosen epoch.

- (b) Compare δ_1 with $\delta_1 + \delta_2$. Measure their power spectra and at least one higher-order statistic: selected bispectrum triangle bins or the smoothed reduced skewness

$$S_3(R) = \frac{\langle \delta_R^3 \rangle}{\langle \delta_R^2 \rangle^2}, \quad \delta_R(\mathbf{k}) = W_R(k) \delta(\mathbf{k}).$$

State the smoothing window and smoothing radius if using S_3 .

- (c) Test the Gaussian higher-order statistic over multiple realizations. A single finite box need not give exactly zero. For a bispectrum measurement, compare the weakly nonlinear ensemble mean with

$$B_{\text{tree}} = 2F_2(\mathbf{k}_1, \mathbf{k}_2) P_L(k_1) P_L(k_2) + \text{two cyclic terms}, \quad \mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0.$$

For equilateral triangles the lecture's target is $B_{\text{tree}}^{\text{equil}} = 12P_L^2/7$. Average the prediction over the same triangle bins used by the estimator.

- (d) Test resolution, box size, realization scatter, and input amplitude. Restrict comparisons to well-resolved modes below the Nyquist scale; explain how the non-Gaussian signal differs from finite-sampling noise.

Supplementary check from the notes: one-loop integration

If implementing the one-loop integrals, use the convention $P_{1\text{-loop}} = P_L + P_{22} + 2P_{13}$. Test logarithmic radial sampling, the region $|\mathbf{k} - \mathbf{q}| \rightarrow 0$, and the stability of $P_{22} + 2P_{13}$ as the infrared cutoff is lowered. Vary the ultraviolet cutoff and check that the result has the same units as P_L . Discuss separately numerical convergence and sensitivity to unresolved short-scale physics. This is supplementary, not part of the minimal higher-order-statistics practical.

Source: Lecture 6, slide 38; notes, Section 6, Section 7.5, and Appendix C.

Exercise 3. Generate and test Lagrangian initial conditions

Lecture 7 — required ZA practical and optional 2LPT extension

Use a periodic cube of side L , an N_g^3 mesh, and a uniform particle load. Declare the input spectrum $P_L(k, a_{\text{ref}})$ and adopt $D_1(a_{\text{ref}}) = 1$, within the pressureless, scale-independent-growth setting of this lecture.

- (a) **Draw a Gaussian field.** Use the periodic-box convention

$$\delta(\mathbf{x}) = \frac{1}{V} \sum_{\mathbf{k}} \delta_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}}, \quad \langle |\delta_{\mathbf{k}}|^2 \rangle = V P_L(k).$$

For a non-self-conjugate pair, draw $\delta_{\mathbf{k}} = \sqrt{V P_L(k)/2} (g_1 + i g_2)$, with independent standard normal g_1, g_2 , and set $\delta_{-\mathbf{k}} = \delta_{\mathbf{k}}^*$. A cell that coincides with its negative-index cell must be real, with variance $V P_L(k)$; set the zero mode to zero. Document the FFT-library conversion and the convention for derivatives on Nyquist components.

- (b) **Construct ZA positions and velocities.** For nonzero modes, compute

$$\Psi_{\mathbf{k}}^{(1)}(a_i) = i D_1(a_i) \frac{\mathbf{k}}{k^2} \delta_{\text{ref}, \mathbf{k}}, \quad \mathbf{x}_p = \mathbf{q}_p + \Psi_p^{(1)}, \quad \mathbf{v}_p = \mathcal{H}_i f_{1,i} \Psi_p^{(1)}.$$

Here $f_1 = d \ln D_1 / d \ln a$, evaluated at a_i for the velocities. Inverse transform, evaluate at the particle labels, and wrap positions into $[0, L)$. Set the mean displacement to zero. If the spectrum is already supplied at a_i , use it directly rather than applying its growth a second time.

- (c) **Validate each step.** Check the mean density, Hermitian symmetry, imaginary FFT residual, displacement curl and divergence, total mass, and velocity normalization. With consistent discrete operators, require $-\nabla \cdot \Psi^{(1)} = D_1 \delta_{\text{ref}}$.
- (d) **Recover the spectrum.** Assign particles to a mesh and measure

$$\hat{P}(k) = \frac{1}{N_k V} \sum_{\mathbf{k}_a \in S_k} |\delta_{\mathbf{k}_a}|^2.$$

Here S_k is a wave-number shell containing N_k Fourier modes. Compare the Gaussian field and ZA particle field at the same epoch. Document the mass-assignment window, particle load, and usable range of k . Do not assume a lattice or glass has Poisson shot noise. Repeat with different seeds and resolutions.

Extension stated in the lecture: 2LPT and matched statistics

Implement the Hessian-product source and the second Poisson solve from the notes. Use the same phases, reference spectrum, and epoch as ZA, and assign velocities with the matching second-order growth rate. Verify that the second-order source vanishes in one dimension, and compare it with a direct convolution on a small mesh. Measure S_3 or selected bispectrum triangles for the Gaussian, ZA, and 2LPT outputs using the estimators of Exercise 2.

Vary z_{ini} . To test *transients*, evolve the different initial conditions with the same evolution solver to a common final epoch and compare statistics there; initial snapshots at different epochs alone do not measure transient decay. Keep all intermediate outputs and normalization metadata.

Source: Lecture 7, slides 47–58; notes, Sections 7–8 and Appendix B.

Exercise 4. One spherical collapse, three numerical formulations

Lecture 8 — required implementation assignment

Implement formulations A, B, and C independently, following the lecture’s smooth-dark-energy, shear-free limit of Pace, Meyer & Bartelmann (2017). Use EdS and flat Λ CDM, with radiation neglected consistently even for the early numerical starting epochs. Define

$$A_H = 2 + \frac{d \ln H}{dN}, \quad \vartheta = \frac{\nabla_x \cdot \mathbf{v}}{aH}, \quad g = \frac{1}{\delta}.$$

Here g is inverse overdensity, not the linear growth rate f .

- (a) **Implement all three systems.** Convert A to a first-order ODE system and integrate B and C directly:

$$\text{A: } \delta_{NN} + A_H \delta_N - \frac{4}{3} \frac{\delta_N^2}{1 + \delta} - \frac{3}{2} \Omega_m \delta (1 + \delta) = 0;$$

$$\text{B: } \delta_N = -(1 + \delta) \vartheta, \quad \vartheta_N = -A_H \vartheta - \frac{\vartheta^2}{3} - \frac{3}{2} \Omega_m \delta;$$

$$\text{C: } g_N = g(1 + g) \vartheta, \quad \vartheta_N = -A_H \vartheta - \frac{\vartheta^2}{3} - \frac{3\Omega_m}{2g}.$$

Check their algebraic equivalence, but do not replace independent implementations by three output transformations of a single integration.

- (b) **Select the growing mode.** For a trial $\delta_i > 0$ at $a_i \ll a_c$, obtain f_i from Exercise 1 and use

$$\delta_{N,i} = f_i \delta_i, \quad \vartheta_i = -\frac{f_i \delta_i}{1 + \delta_i}, \quad g_i = \delta_i^{-1}.$$

Integrate the linear counterpart in parallel,

$$\delta_{L,NN} + A_H \delta_{L,N} - \frac{3}{2} \Omega_m \delta_L = 0, \quad \delta_{L,i} = \delta_i, \quad \delta_{L,N,i} = f_i \delta_i.$$

- (c) **Shoot for the target collapse epoch.** For $a_c = (1 + z_c)^{-1}$, bracket and tune δ_i until the nonlinear terminal event occurs at a_c . Use $\delta = \delta_\infty$ for A and B, and $g = \epsilon > 0$ for C. Read $\delta_c = \delta_L(a_c)$ from the corresponding linear solution. Test the finite-threshold limit; never integrate across the singular endpoint. In particular, C still contains the singular term $1/g$ and a divergent velocity variable.

Required outputs from each formulation: the tuned initial amplitude, turnaround epoch and density, δ_c , and both virial-overdensity conventions defined on the following page. Record root-search brackets, event residuals, and solver settings.

Source: Lecture 8, slides 51–58; notes, Sections 6–7. Pace, Meyer & Bartelmann, “On the implementation of the spherical collapse model for dark energy models,” JCAP 10, 040 (2017), [arXiv:1708.02477](https://arxiv.org/abs/1708.02477).

Exercise 4 continued: turnaround, virialization, and convergence

- (d) **Locate turnaround and apply a common virial prescription.** Find $\vartheta_{\text{ta}} = -3$ and $\zeta_{\text{ta}} = 1 + \delta(a_{\text{ta}})$. For A reconstruct $\vartheta = -\delta_N/(1 + \delta)$; for C reconstruct $\delta = 1/g$. Let

$$y = \frac{R_{\text{vir}}}{R_{\text{ta}}}, \quad \lambda_{\text{ta}} = \frac{\rho_{\Lambda}}{\rho_{\text{sph,ta}}}.$$

Use $y = 1/2$ in EdS. In flat Λ CDM solve

$$4\lambda_{\text{ta}}y^3 - 2(1 + \lambda_{\text{ta}})y + 1 = 0$$

for the physical root connected continuously to $1/2$ as $\lambda_{\text{ta}} \rightarrow 0$. Adopt the *formal-collapse epoch* convention:

$$\Delta_{\text{vir},m}(a_c) = \zeta_{\text{ta}} \left(\frac{a_c}{a_{\text{ta}}} \right)^3 y^{-3}, \quad \Delta_{\text{vir},c}(a_c) = \Omega_m(a_c) \Delta_{\text{vir},m}(a_c).$$

The subscripts m and c indicate mean-matter and critical reference density; the subscript of δ_c instead denotes the collapse threshold.

- (e) **Recover independent EdS benchmarks.** At more than one collapse redshift, test

$$\delta_c \simeq 1.68647, \quad \frac{a_{\text{ta}}}{a_c} = 2^{-2/3}, \quad \zeta_{\text{ta}} = \left(\frac{3\pi}{4} \right)^2, \quad \Delta_{\text{vir},m} = 18\pi^2.$$

Do not tune an event threshold to reproduce only δ_c while the turnaround checks fail.

- (f) **Run the prescribed cosmology grid.** Use

$$\Omega_{m0} \in \{0.2, 0.3, 0.4\}, \quad \Omega_{\Lambda 0} = 1 - \Omega_{m0}, \quad z_c \in \{0, 1, 2\}.$$

Plot $\delta_c(z_c)$, $\Delta_{\text{vir},m}(z_c)$, and $\Delta_{\text{vir},c}(z_c)$, with EdS baselines. Explain the separate roles of growth, turnaround, the virial prescription, and reference density.

- (g) **Demonstrate convergence.** Compare $a_i = 10^{-3}, 10^{-4}, 10^{-5}$, several relative and absolute tolerances, and $\delta_{\infty} = 10^5, 10^6, 10^7, 10^8$ or $\epsilon = 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}$. Retune δ_i when changing these settings. Tabulate cross-formulation differences and event residuals; discuss the numerical failure modes of A, B, and C separately.

Submission specified in Lecture 8. Supply runnable code for all three formulations, a README with dependencies and execution commands, and scripts or saved data reproducing the figures. Include a LaTeX report (`.tex` and PDF) with equations, initial conditions, virial conventions, benchmark and convergence tables, cosmology plots, and references. Implementing a halo mass function or fitting a new halo-abundance calibration is *not* required.

Exercise 5. Validate the primordial-to-late-time numerical map

Lecture 9 — solver validation from the slides and accompanying notes

Use an existing Boltzmann solver; writing a new Einstein–Boltzmann code is not part of this task. The supplied CLASS configuration is `09-Primordial-Boltzmann/Figs/Scripts/class_base.ini`. Keep a separate working copy and output directory. Its accompanying scripts and archived tables provide the lecture’s reference calculation.

- (a) **Reconstruct the matter spectrum.** Run the declared fiducial model with nonlinear matter corrections disabled. Record the code version, all physical parameters, output redshift, gauge, matter component, and units. Convert the density transfer to $T_m = \delta_m/\mathcal{R}$ and reconstruct

$$P_m^{\text{rec}}(k, z) = \frac{2\pi^2}{k^3} A_s \left(\frac{k}{k_*} \right)^{n_s-1} T_m^2(k, z).$$

Compare with the solver’s direct linear P_m on the same k grid. For tabulated numbers k_h in $h \text{ Mpc}^{-1}$ and P_h in $(\text{Mpc}/h)^3$, use $k = hk_h$ in Mpc^{-1} and $P = P_h/h^3$ in Mpc^3 . Plot the fractional reconstruction residual.

- (b) **Separate identity checks from convergence.** Explain why the reconstruction tests normalization and units but not the solver’s physical accuracy. Obtain spectra with tighter precision settings and plot ratios to that reference. Vary numerical controls one at a time: ODE tolerances, wave-number and time sampling, hierarchy depth or closure, approximation-switch thresholds, maximum multipole, and massive-species momentum sampling where applicable. Record the actual controls exposed by the chosen code.
- (c) **Check independent implementations.** Compare CLASS and CAMB only after matching physical parameters, neutrino and recombination models, units, gauge and transfer normalization, matter component, linear/nonlinear choice, and requested precision. Separate changes in physical modelling (for example a recombination model) from changes in numerical tolerance.

Supplementary check A: collisionless-hierarchy closure

For the source-free massless hierarchy, use its spherical-Bessel solution as a test. Compare an outgoing closure with simply setting the next multipole to zero. Increase ℓ_{max} until the retained low multipoles are stable, and diagnose the reflection of angular power caused by a premature truncation.

Source: Lecture 9, backup slide 59; notes, Appendix B.

Supplementary check B: the radiation-to-matter potential ratio

In the idealized superhorizon, adiabatic, negligible-anisotropic-stress limit, set $y = a/a_{\text{eq}}$, $\mathcal{R} = 1$, and integrate in $\ln y$:

$$\frac{d\phi}{d \ln y} + \frac{6 + 5y}{2(1 + y)} \phi = -\frac{4 + 3y}{2(1 + y)}.$$

Start at small y_i with $\phi_i = -2/3 + y_i/24$. Recover the matter-era limit $-3/5$, verify the amplitude ratio $9/10$, and reconstruct the conserved curvature using Appendix A of the notes. Test the initial epoch and tolerances. Explain why this is not a universal suppression factor for P_m .

Source: Lecture 9, slides 52–53; notes, Sections 8.4–8.6 and Appendices A–C. The supplementary checks are not additional required implementation projects.